

## SAFETY DATA SHEET

Version 6.11  
Revision Date 09/08/2024  
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**SECTION 1: Identification of the substance/mixture and of the company/undertaking****1.1 Product identifiers**

Product name : Tri(propylene glycol) diacrylate, mixture of isomers

Product Number : 246832  
Brand : Aldrich  
Index-No. : 607-249-00-X  
CAS-No. : 42978-66-5

**1.2 Relevant identified uses of the substance or mixture and uses advised against**

Identified uses : Laboratory chemicals, Synthesis of substances

Uses advised against : The product is being supplied under the TSCA R&D Exemption (40 CFR Section 720.36). It is the recipient's responsibility to comply with the requirements of the R&D exemption. The product may not be used for a non-exempt commercial purpose under TSCA unless appropriate consent is granted in writing by MilliporeSigma.

**1.3 Details of the supplier of the safety data sheet**

Company : Sigma-Aldrich Inc.  
3050 SPRUCE ST  
ST. LOUIS MO 63103  
UNITED STATES

Telephone : +1 314 771-5765  
Fax : +1 800 325-5052

**1.4 Emergency telephone**

Emergency Phone # : 800-424-9300 CHEMTREC (USA) +1-703-527-3887 CHEMTREC (International) 24 Hours/day; 7 Days/week

**SECTION 2: Hazards identification****2.1 Classification of the substance or mixture****GHS Classification in accordance with 29 CFR 1910 (OSHA HCS)**

Skin irritation (Category 2), H315  
Eye irritation (Category 2A), H319

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Skin sensitization (Category 1), H317

Specific target organ toxicity - single exposure (Category 3), Respiratory system, H335

Short-term (acute) aquatic hazard (Category 2), H401

Long-term (chronic) aquatic hazard (Category 2), H411

For the full text of the H-Statements mentioned in this Section, see Section 16.

## 2.2 GHS Label elements, including precautionary statements

Pictogram



Signal Word

Warning

Hazard Statements

H315

Causes skin irritation.

H317

May cause an allergic skin reaction.

H319

Causes serious eye irritation.

H335

May cause respiratory irritation.

H411

Toxic to aquatic life with long lasting effects.

Precautionary Statements

P261

Avoid breathing mist or vapors.

P264

Wash skin thoroughly after handling.

P271

Use only outdoors or in a well-ventilated area.

P272

Contaminated work clothing must not be allowed out of the workplace.

P273

Avoid release to the environment.

P280

Wear protective gloves/ eye protection/ face protection.

P302 + P352

IF ON SKIN: Wash with plenty of soap and water.

P304 + P340 + P312

IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER/ doctor if you feel unwell.

P305 + P351 + P338

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P333 + P313

If skin irritation or rash occurs: Get medical advice/ attention.

P337 + P313

If eye irritation persists: Get medical advice/ attention.

P362

Take off contaminated clothing and wash before reuse.

P391

Collect spillage.

P403 + P233

Store in a well-ventilated place. Keep container tightly closed.

P405

Store locked up.

P501

Dispose of contents/ container to an approved waste disposal plant.

## 2.3 Hazards not otherwise classified (HNOC) or not covered by GHS - none

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### SECTION 3: Composition/information on ingredients

#### 3.1 Substances

Synonyms : TRPGDA

Formula :  $C_{15}H_{24}O_6$

Molecular weight : 300.35 g/mol

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CAS-No. : 42978-66-5  
 EC-No. : 256-032-2  
 Index-No. : 607-249-00-X

| Component                                                   | Classification                                                                                                                                                                                     | Concentration |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Tri(propylene glycol) diacrylate, mixture of isomers</b> |                                                                                                                                                                                                    |               |
|                                                             | Skin Irrit. 2; Eye Irrit. 2A;<br>Skin Sens. 1; STOT SE 3;<br>Aquatic Acute 2; Aquatic<br>Chronic 2; H315, H319,<br>H317, H335, H401, H411<br>Concentration limits:<br>>= 10 %: STOT SE 3,<br>H335; | <= 100 %      |

For the full text of the H-Statements mentioned in this Section, see Section 16.

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## SECTION 4: First aid measures

### 4.1 Description of first-aid measures

#### General advice

Show this material safety data sheet to the doctor in attendance.

#### If inhaled

After inhalation: fresh air.

#### In case of skin contact

In case of skin contact: Take off immediately all contaminated clothing. Rinse skin with water/ shower. Consult a physician.

#### In case of eye contact

After eye contact: rinse out with plenty of water. Call in ophthalmologist. Remove contact lenses.

#### If swallowed

After swallowing: caution if victim vomits. Risk of aspiration! Keep airways free. Pulmonary failure possible after aspiration of vomit. Call a physician immediately.

### 4.2 Most important symptoms and effects, both acute and delayed

The most important known symptoms and effects are described in the labelling (see section 2.2) and/or in section 11

### 4.3 Indication of any immediate medical attention and special treatment needed

No data available

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## SECTION 5: Firefighting measures

### 5.1 Extinguishing media

#### Suitable extinguishing media

Water Foam Carbon dioxide (CO2) Dry powder

**Unsuitable extinguishing media**

For this substance/mixture no limitations of extinguishing agents are given.

**5.2 Special hazards arising from the substance or mixture**

Carbon oxides

Combustible.

Vapors are heavier than air and may spread along floors.

Forms explosive mixtures with air on intense heating.

Development of hazardous combustion gases or vapours possible in the event of fire.

**5.3 Advice for firefighters**

Stay in danger area only with self-contained breathing apparatus. Prevent skin contact by keeping a safe distance or by wearing suitable protective clothing.

**5.4 Further information**

Prevent fire extinguishing water from contaminating surface water or the ground water system.

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**SECTION 6: Accidental release measures****6.1 Personal precautions, protective equipment and emergency procedures**

Advice for non-emergency personnel: Do not breathe vapors, aerosols. Avoid substance contact. Ensure adequate ventilation. Evacuate the danger area, observe emergency procedures, consult an expert.

For personal protection see section 8.

**6.2 Environmental precautions**

Do not let product enter drains.

**6.3 Methods and materials for containment and cleaning up**

Cover drains. Collect, bind, and pump off spills. Observe possible material restrictions (see sections 7 and 10). Take up with liquid-absorbent material (e.g. Chemizorb® ). Dispose of properly. Clean up affected area.

**6.4 Reference to other sections**

For disposal see section 13.

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**SECTION 7: Handling and storage****7.1 Precautions for safe handling**

For precautions see section 2.2.

**7.2 Conditions for safe storage, including any incompatibilities****Storage conditions**

Tightly closed.

Light sensitive. Hygroscopic.

**Storage class**

Storage class (TRGS 510): 10: Combustible liquids

**7.3 Specific end use(s)**

Apart from the uses mentioned in section 1.2 no other specific uses are stipulated

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## SECTION 8: Exposure controls/personal protection

### 8.1 Control parameters

#### Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

### 8.2 Exposure controls

#### Appropriate engineering controls

Immediately change contaminated clothing. Apply preventive skin protection. Wash hands and face after working with substance.

#### Personal protective equipment

##### Eye/face protection

Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU). Safety glasses

##### Skin protection

required

##### Body Protection

protective clothing

##### Respiratory protection

Recommended Filter type: Filter type ABEK

The entrepreneur has to ensure that maintenance, cleaning and testing of respiratory protective devices are carried out according to the instructions of the producer.

These measures have to be properly documented.

required when vapours/aerosols are generated.

Our recommendations on filtering respiratory protection are based on the following standards: DIN EN 143, DIN 14387 and other accompanying standards relating to the used respiratory protection system.

#### Control of environmental exposure

Do not let product enter drains.

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## SECTION 9: Physical and chemical properties

### 9.1 Information on basic physical and chemical properties

- |                                 |                                                   |
|---------------------------------|---------------------------------------------------|
| a) Appearance                   | Form: liquid<br>Color: light yellow               |
| b) Odor                         | No data available                                 |
| c) Odor Threshold               | No data available                                 |
| d) pH                           | No data available                                 |
| e) Melting point/freezing point | Melting point/ range: < -20 °C (< -4 °F) - (ECHA) |

|                                                 |                                                                                                                                           |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| f) Initial boiling point and boiling range      | > 120 °C > 248 °F at 1,013.25 hPa - (ECHA)                                                                                                |
| g) Flash point                                  | 153 °C (307 °F) - closed cup - ISO 2719                                                                                                   |
| h) Evaporation rate                             | No data available                                                                                                                         |
| i) Flammability (solid, gas)                    | No data available                                                                                                                         |
| j) Upper/lower flammability or explosive limits | No data available                                                                                                                         |
| k) Vapor pressure                               | < 0.01 hPa at 20 °C (68 °F)                                                                                                               |
| l) Vapor density                                | 10.36 - (Air = 1.0)                                                                                                                       |
| m) Density                                      | 1.03 g/cm <sup>3</sup> at 25 °C (77 °F) - lit.                                                                                            |
| Relative density                                | No data available                                                                                                                         |
| n) Water solubility                             | 4 g/l at 20 °C (68 °F) - OECD Test Guideline 105                                                                                          |
| o) Partition coefficient: n-octanol/water       | log Pow: > 2.5 - < 2.7 at 23 °C (73 °F) - Bioaccumulation is not expected.                                                                |
| p) Autoignition temperature                     | 214 °C (417 °F) at 1,000 hPa                                                                                                              |
| q) Decomposition temperature                    | No data available                                                                                                                         |
| r) Viscosity                                    | 14.5 mm <sup>2</sup> /s at 20 °C (68 °F) - OECD Test Guideline 114 - 6.7 mm <sup>2</sup> /s at 40 °C (104 °F) - OECD Test Guideline 114 - |
| s) Explosive properties                         | No data available                                                                                                                         |
| t) Oxidizing properties                         | none                                                                                                                                      |

## 9.2 Other safety information

|                        |                     |
|------------------------|---------------------|
| Relative vapor density | 10.36 - (Air = 1.0) |
|------------------------|---------------------|

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## SECTION 10: Stability and reactivity

### 10.1 Reactivity

Forms explosive mixtures with air on intense heating.  
A range from approx. 15 Kelvin below the flash point is to be rated as critical.

### 10.2 Chemical stability

The product is chemically stable under standard ambient conditions (room temperature) .

### 10.3 Possibility of hazardous reactions

No data available

### 10.4 Conditions to avoid

May polymerize on exposure to light. Avoid moisture.  
Strong heating.

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## 10.5 Incompatible materials

Oxidizing agents Strong acids, Strong oxidizing agents, Strong bases, Brass, Copper, Steel (all types and surface treatments), Iron and iron salts.

## 10.6 Hazardous decomposition products

In the event of fire: see section 5

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## SECTION 11: Toxicological information

### 11.1 Information on toxicological effects

#### Acute toxicity

LD50 Oral - Rat - female - > 2,000 mg/kg (Tri(propylene glycol) diacrylate, mixture of isomers)

(OECD Test Guideline 423)

Inhalation: No data available

LD50 Dermal - Rabbit - male and female - > 2,000 mg/kg (Tri(propylene glycol) diacrylate, mixture of isomers)

(OECD Test Guideline 402)

No data available

#### Skin corrosion/irritation

Remarks: Causes skin irritation.

Classified according to Regulation (EU) 1272/2008, Annex VI (Table 3.1/3.2)

#### Serious eye damage/eye irritation

Eyes - Rabbit (Tri(propylene glycol) diacrylate, mixture of isomers)

Result: Mild eye irritation - 24 h

(OECD Test Guideline 405)

Remarks: Causes serious eye irritation.

Classified according to Regulation (EU) 1272/2008, Annex VI (Table 3.1/3.2)

#### Respiratory or skin sensitization

Local lymph node assay (LLNA) - Mouse (Tri(propylene glycol) diacrylate, mixture of isomers)

Result: positive

(OECD Test Guideline 429)

May cause an allergic skin reaction. Classified according to Regulation (EU) 1272/2008, Annex VI (Table 3.1/3.2) (Tri(propylene glycol) diacrylate, mixture of isomers)

#### Germ cell mutagenicity

Test Type: gene mutation test

(Tri(propylene glycol) diacrylate, mixture of isomers)

Test system: Chinese hamster ovary cells

Metabolic activation: with and without metabolic activation

Method: OECD Test Guideline 476

Result: negative

Test Type: In vitro mammalian cell gene mutation test

(Tri(propylene glycol) diacrylate, mixture of isomers)

Test system: mouse lymphoma cells

Metabolic activation: with and without metabolic activation

Result: positive

Remarks: (ECHA)

(Tri(propylene glycol) diacrylate, mixture of isomers)  
Test Type: In vivo micronucleus test  
Species: Mouse

Application Route: Intraperitoneal  
Method: OECD Test Guideline 474  
Result: negative

### **Carcinogenicity**

IARC: No ingredient of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.

NTP: No ingredient of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP.

OSHA: No component of this product present at levels greater than or equal to 0.1% is on OSHA's list of regulated carcinogens.

### **Reproductive toxicity**

No data available

### **Specific target organ toxicity - single exposure**

May cause respiratory irritation. - Respiratory system (Tri(propylene glycol) diacrylate, mixture of isomers)

Remarks: Classified according to Regulation (EU) 1272/2008, Annex VI (Table 3.1/3.2)

### **Specific target organ toxicity - repeated exposure**

No data available

### **Aspiration hazard**

No data available

## **11.2 Additional Information**

Repeated dose toxicity - Rat - male and female - Oral - NOAEL (No observed adverse effect level) - 375 mg/kg

(Tri(propylene glycol) diacrylate, mixture of isomers)

Repeated dose toxicity - Rat - male and female - Dermal - 90 d - LOAEL (Lowest observed adverse effect level) - 20 mg/kg

(Tri(propylene glycol) diacrylate, mixture of isomers)

RTECS: AT4690000

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated. (Tri(propylene glycol) diacrylate, mixture of isomers)

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## **SECTION 12: Ecological information**

### **12.1 Toxicity**

|                                       |                                                                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Toxicity to fish                      | static test LC50 - Leuciscus idus (Golden orfe) - > 4.6 - < 10 mg/l - 96 h (Tri(propylene glycol) diacrylate, mixture of isomers) (DIN 38412) |
| Toxicity to daphnia and other aquatic | static test EC50 - Daphnia magna Straus (Water flea) - 89 mg/l - 48 h (Tri(propylene glycol) diacrylate, mixture of isomers)                  |

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|                      |                                                                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| invertebrates        | (Regulation (EC) No. 440/2008, Annex, C.2)                                                                                                              |
| Toxicity to algae    | static test ErC50 - <i>Desmodesmus subspicatus</i> (green algae) - 65.9 mg/l - 72 h (Tri(propylene glycol) diacrylate, mixture of isomers) (DIN 38412)  |
| Toxicity to bacteria | Respiration inhibition EC50 - Sludge Treatment - > 1,000 mg/l - 30 min (Tri(propylene glycol) diacrylate, mixture of isomers) (OECD Test Guideline 209) |

## 12.2 Persistence and degradability

|                  |                                                                                                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Biodegradability | aerobic - Exposure time 28 d (Tri(propylene glycol) diacrylate, mixture of isomers)<br>Result: 48 % - Partially biodegradable.<br>(OECD Test Guideline 301B) |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 12.3 Bioaccumulative potential

No data available

## 12.4 Mobility in soil

No data available

## 12.5 Results of PBT and vPvB assessment

PBT/vPvB assessment not available as chemical safety assessment not required/not conducted

## 12.6 Endocrine disrupting properties

No data available

## 12.7 Other adverse effects

No data available

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## SECTION 13: Disposal considerations

### 13.1 Waste treatment methods

#### Product

Waste material must be disposed of in accordance with the national and local regulations. Leave chemicals in original containers. No mixing with other waste. Handle uncleaned containers like the product itself.

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## SECTION 14: Transport information

### DOT (US)

Not dangerous goods

### IMDG

UN number: 3082 Class: 9

Packing group: III

EMS-No: F-A, S-F

Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.  
(Tri(propylene glycol) diacrylate, mixture of isomers)

Marine pollutant : yes

Marine pollutant : no

**IATA**

UN number: 3082 Class: 9

Packing group: III

Proper shipping name: Environmentally hazardous substance, liquid, n.o.s. (Tri(propylene glycol) diacrylate, mixture of isomers)

**Further information**

EHS-Mark required (ADR 2.2.9.1.10, IMDG code 2.10.3) for single packagings and combination packagings containing inner packagings with Dangerous Goods > 5L for liquids or > 5kg for solids. Packages smaller than or equal to 5 kg / L , not dangerous goods of Class 9

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## SECTION 15: Regulatory information

### CERCLA Reportable Quantity

Listed substances in the product are at low enough levels to not be expected to exceed the RQ

### SARA 304 Extremely Hazardous Substances Reportable Quantity

Listed substances in the product are at low enough levels to not be expected to exceed the RQ

### SARA 302 Extremely Hazardous Substances Threshold Planning Quantity

This material does not contain any components with a section 302 EHS TPQ.

### SARA 313

: This material does not contain any chemical components with known CAS numbers that exceed the threshold (De Minimis) reporting levels established by SARA Title III, Section 313.

## US State Regulations

### Massachusetts Right To Know

hydroquinone

123-31-9

### Maine Chemicals of High Concern

Product does not contain any listed chemicals

### Vermont Chemicals of High Concern

Product does not contain any listed chemicals

### Washington Chemicals of High Concern

Product does not contain any listed chemicals

## The ingredients of this product are reported in the following inventories:

TSCA : All substances listed as active on the TSCA inventory

### TSCA list

No substances are subject to a Significant New Use Rule.

No substances are subject to TSCA 12(b) export notification requirements.

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## SECTION 16: Other information

### Further information

The information is believed to be correct but is not exhaustive and will be used solely as a guideline, which is based on current knowledge of the chemical substance or mixture and is applicable to appropriate safety precautions for the product. It does not represent any guarantee of the properties of the product. Sigma-Aldrich Corporation and its Affiliates shall not be held liable for any damage resulting from handling or from contact with the above product. See [www.sigma-aldrich.com](http://www.sigma-aldrich.com) and/or the reverse side of invoice or packing slip for additional terms and conditions of sale.

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